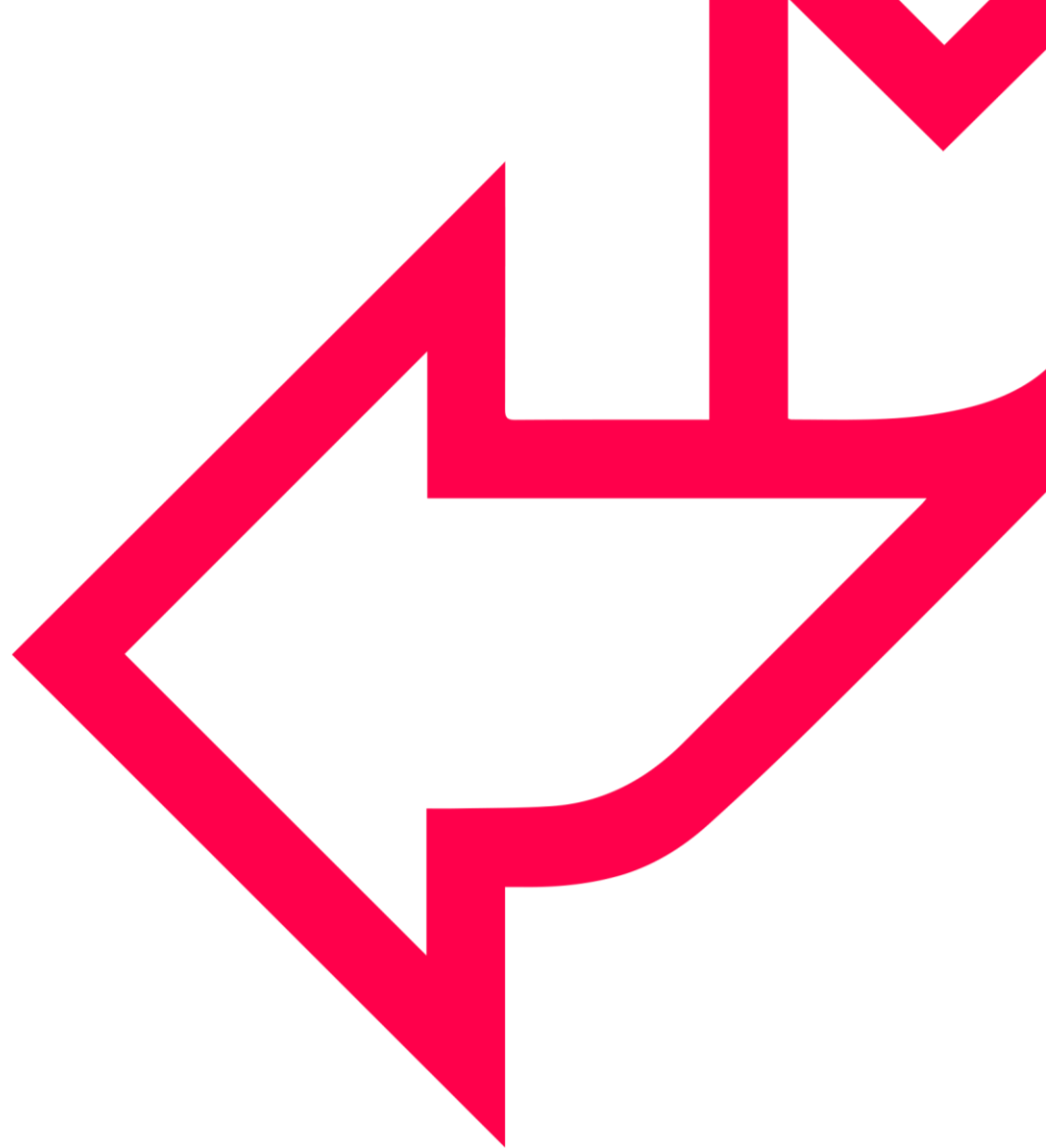




File I/O





READ THE FIRST LINE OF A FILE

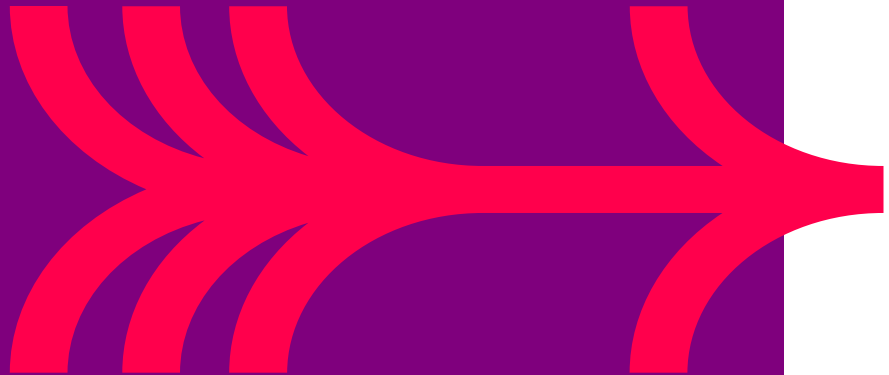
```
file = open("students.txt", "r")  
line = file.readline()
```

```
file.close()
```

```
print(line)
```

```
FirstName, Surname, Math, English, IT, Biology, PE
```

```
FirstName, Surname, Math, English, IT, Biology, PE  
Alfalfa, Aloysius, 40.0, 90.0, 100.0, 83.0, 49.0  
Alfred, Smith, 41.0, 97.0, 96.0, 97.0, 48.0
```





READ EVERY LINE OF A FILE INTO MEMORY

```
file = open("students.txt", "r")  
lines = file.readlines()
```

```
file.close()
```

```
for line in lines:  
    print(line)
```

lines	
0	'FirstName,Surname,Math,English,IT,Biology,PE\n'
1	'Alfalfa,Aloysius,40.0,90.0,100.0,83.0,49.0\n'
2	'Alfred,Smith,41.0,97.0,96.0,97.0,48.0\n'
__len__	3

```
FirstName, Surname, Math, English, IT, Biology, PE  
Alfalfa, Aloysius, 40.0, 90.0, 100.0, 83.0, 49.0  
Alfred, Smith, 41.0, 97.0, 96.0, 97.0, 48.0
```

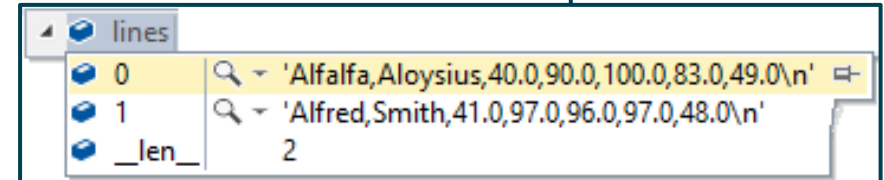


READ EVERY LINE OF A FILE INTO MEMORY

```
file = open("students.txt", "r")  
header = file.readline()  
lines = file.readlines()
```

```
file.close()
```

```
for line in lines:  
    print(line)
```



lines	
0	'Alfalfa,Aloysius,40.0,90.0,100.0,83.0,49.0\n'
1	'Alfred,Smith,41.0,97.0,96.0,97.0,48.0\n'
__len__	2

```
FirstName, Surname, Math, English, IT, Biology, PE  
Alfalfa, Aloysius, 40.0, 90.0, 100.0, 83.0, 49.0  
Alfred, Smith, 41.0, 97.0, 96.0, 97.0, 48.0
```



READ A FILE LINE BY LINE

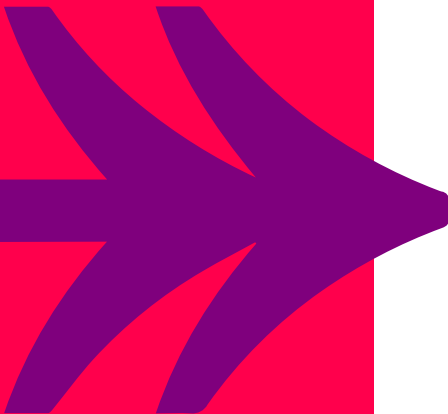
```
file = open("students.txt", "r")
```

```
for line in file:  
    print(line.strip())
```

```
file.close()
```

```
FirstName, Surname, Math, English, IT, Biology, PE  
Alfalfa, Aloysius, 40.0, 90.0, 100.0, 83.0, 49.0  
Alfred, Smith, 41.0, 97.0, 96.0, 97.0, 48.0  
Press any key to continue ...
```

```
FirstName, Surname, Math, English, IT, Biology, PE  
Alfalfa, Aloysius, 40.0, 90.0, 100.0, 83.0, 49.0  
Alfred, Smith, 41.0, 97.0, 96.0, 97.0, 48.0
```



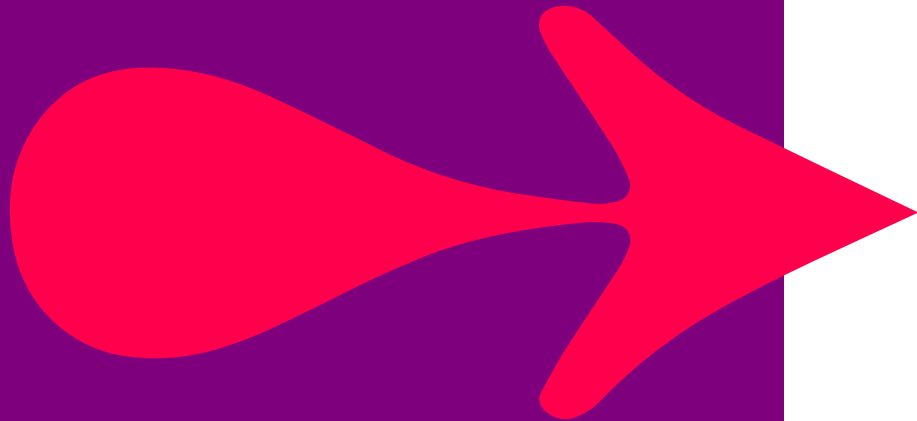


READ ALL LINES INTO MEMORY – USING WITH

```
with open("students.txt") as file:  
    lines = file.readlines()
```

```
for line in lines:  
    print(line)
```

```
FirstName, Surname, Math, English, IT, Biology, PE  
Alfalfa, Aloysius, 40.0, 90.0, 100.0, 83.0, 49.0  
Alfred, Smith, 41.0, 97.0, 96.0, 97.0, 48.0
```





READ A FILE LINE BY LINE – USING WITH

```
with open("students.txt") as file:  
    for line in file:  
        print(line)
```

```
FirstName, Surname, Math, English, IT, Biology, PE  
Alfalfa, Aloysius, 40.0, 90.0, 100.0, 83.0, 49.0  
Alfred, Smith, 41.0, 97.0, 96.0, 97.0, 48.0
```





APPEND TO A FILE – USING WITH

```
with open("students.txt", "a") as file:  
    file.write("Bob, Smith, 30, 80, 70, 60\n")  
    file.write("Rob, Jones, 20, 60, 80, 90\n")
```

FirstName, Surname, Math, English, IT, Biology, PE
Alfalfa, Aloysius, 40.0, 90.0, 100.0, 83.0, 49.0
Alfred, Smith, 41.0, 97.0, 96.0, 97.0, 48.0
Bob, Smith, 30, 80, 70, 60
Rob, Jones, 20, 60, 80, 90



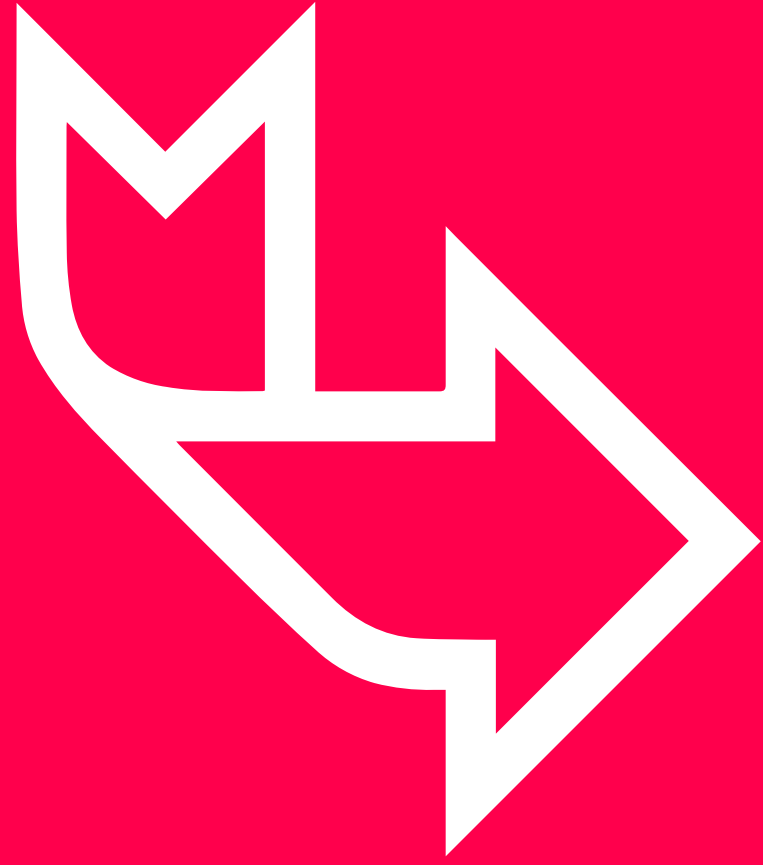
WRITE TO A FILE

```
with open(r"c:\qa\students.txt", "w") as file:  
    file.write("Bob, Smith, 30, 80, 70, 60")
```

```
Bob, Smith, 30, 80, 70, 60
```

The only line in the file!





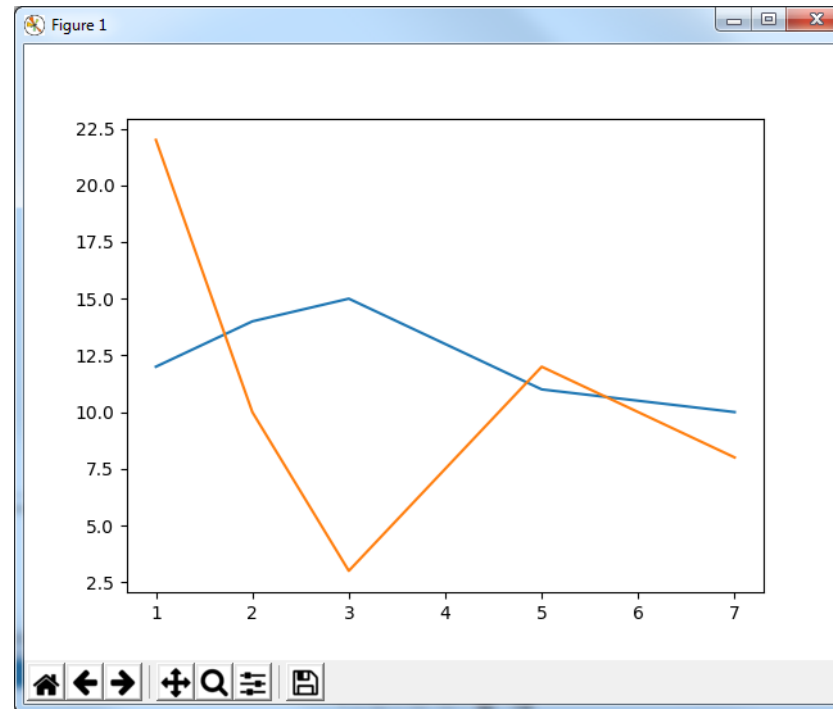
Any questions?

PLOTTING VALUES READ FROM A FILE

```
import matplotlib.pyplot as plt
```

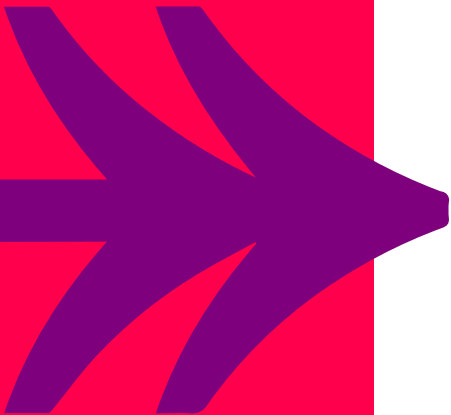
```
plt.plot([1, 2, 3, 5, 7], [12, 14, 15, 11, 10])  
plt.plot([1, 2, 3, 5, 7], [22, 10, 3, 12, 8])
```

```
plt.show()
```





PRACTICAL EXAMPLE OF A GRAPH FROM A FILE



```
import matplotlib.pyplot as plt

with open("students.txt") as file:
    for line in file:
        grades = line.replace(' ', '').split(',')
        grades = list(map(float, grades[2:7]))
        plt.plot([1, 2, 3, 4, 5, ], grades)

plt.show()
```



Thank you!

Hope you enjoyed this learning journey!

